BABEȘ-BOLYAI UNIVERSITY

SUDOKU GLADIATORS - MULTIPLAYER SUDOKU GAME USING FLUTTER

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OVERVIEW

WHAT?	WHY?	HOW?
• DESCRIPTION • MULTIPLAYER FEATURES	• MOTIVATION • EXITING IMPLEMTATIONS • DIFFERENCES FROM COMPETITORS	• TECHNOLOGIES USED • SUDOKU ALGORITHM • IMPLEMENTATION

WHAT IS IT?

Sudoku Gladiators aims to take the Sudoku single-player experince and bring in into the 21st century.

- Ranked queue with Elo based matchmaking
- Casual game modes such as:
  - 1 v 1 on the same Sudoku board
  - co-op mode
  - powerup mode

WHY DID I TAKE ON THIS PROJECT?

- Personal fondness for solving Sudokus since I was just a kid.
- Desire to play together with my friends even if we are far away.
- Curiosity about how multi-player games deal with synchronization issues.

EXISTING IMPLEMENTATIONS

- Web apps:
  - UsDoku - 1v1 with lobby system
  - SudokuBattles - powerups with lobby system

- Academic research on this topic:
  - SudoDuel
    - Powerup system - windows only app developed in Godot
  - Clash of Sudokers
    - 1v1 system - react native with node.js backend on aws

UNIQUENESS

Serverless approach

We do not have a custom node.js server with websockets, we only use firebase and transactions to make the multi-player experience

Cross-platform

The ability to play cross-platform is a founding principle for the development of this thesis and app.

Powerup system

The integration of powerups at random intervals that players have to fight for and specific powerups that affect the other player's board

TECHNOLOGIES

FLUTTER / DART

Flutter enables building natively compiled applications for mobile, web, and desktop from a single codebase. Uses Dart programming language with hot reload for rapid development cycles.

FIREBASE

Google's comprehensive cloud platform providing authentication, real-time database, and cloud storage without server infrastructure management.

IMPLEMENTATION

PROVIDERS

- We have 4 change-notifier providers:
  - Sudoku
  - Powerup
  - Theme
  - Lobby

FIREBASE

- We take advantage of transactions for:
  - Multiplayer sync
  - Claiming victory
  - Much more

SUDOKU ALGORITHM

01

- Uses randomized backtracking to fill empty grid
- Shuffles candidates [1-9] at each cell for variety
- Applies Sudoku constraints (row, column, 3x3 box validation)

02

- Randomly removes cells based on difficulty (35-55 cells)
- For each removal: temporarily delete cell, check if exactly one solution remains
- If unique solution: keep removed; if multiple solutions: restore cell
- Guarantees solvable puzzles with single solutions

CONCLUSION

- Did we manage to implement what we set out to do?
- Is this a feasible approach?

FUTURE WORK

Audio Effects

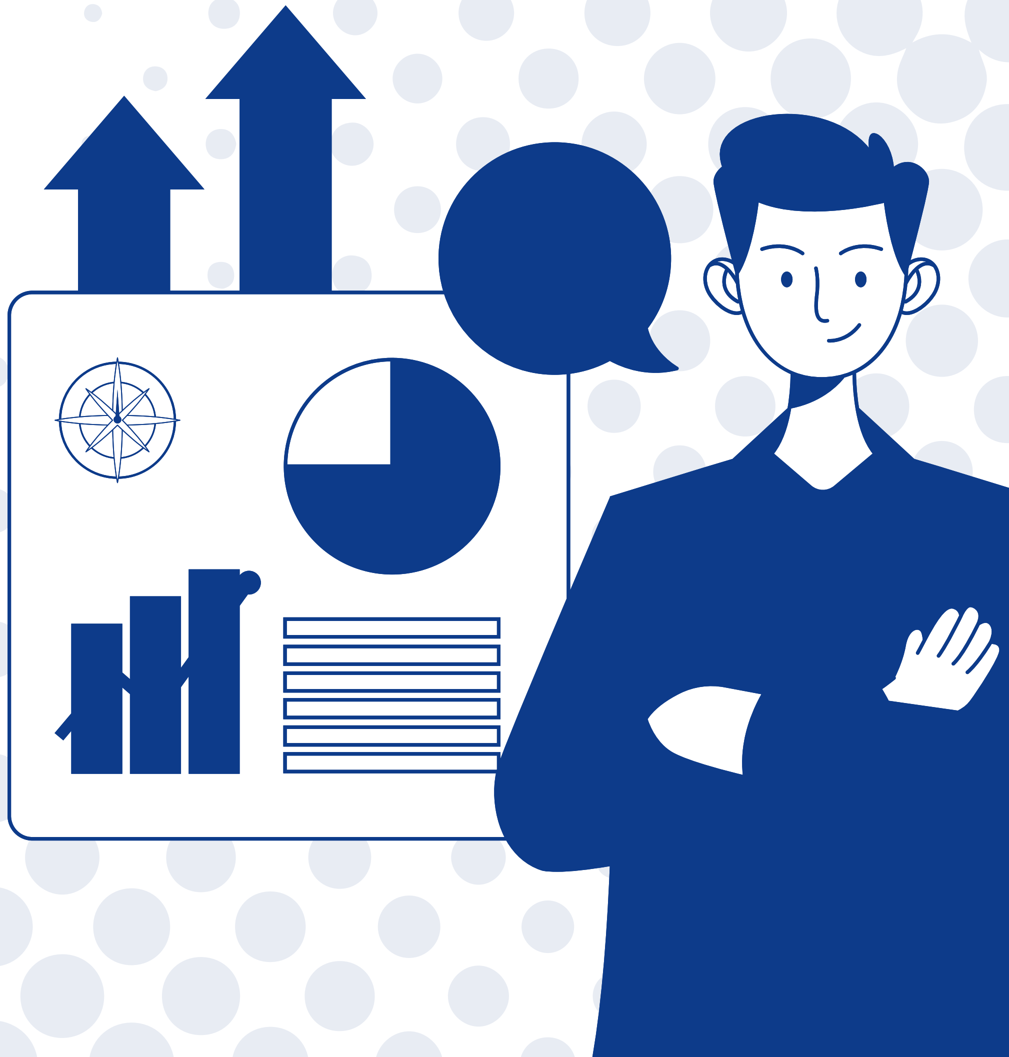
Custom Themes

Friend System

Tournaments

AI for practice and suggestions

Bug Fixing



DEMO

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THANK YOU

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